

Evaluation of PERCLOS based Current Fatigue Monitoring Technologies

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Abstract—In an overnight driving simulation study three commercially available devices of fatigue monitoring technologies (FMT) were applied to test their accuracy. 16 volunteers performed driving tasks during eight sessions (40 min each) separated by 15 minutes breaks. The main output variable of FMT devices, which is the percentage of eye closure (PERCLOS), the driving performance (standard deviation of lateral position in lane, SDL), the electroencephalogram (EEG) and electrooculogram (EOG) were recorded during driving. In addition, the subjective self-rated Karolinska sleepiness scale (KSS) was assessed every 2 min. As expected, Pearson product-moment correlation coefficient (PMCC) yielded significant linear dependence between KSS and PERCLOS as well as between SDL and PERCLOS. However, if PMCC was estimated within smaller data segments (3 min) as well as without averaging across subjects then strongly decreased correlation coefficients resulted. To further validate PERCLOS at higher temporal resolution its ability to discriminate between mild and strong fatigue was investigated and compared to the results of the same analysis for EEG/EOG. Spectral-domain features of both types of signals were classified using Support-Vector Machines (SVM). Results suggest that EEG/EOG indicate driver fatigue much better than PERCLOS. Therefore, current FMT devices perform acceptably if temporal resolution is low (> 20 min). But, even under laboratory conditions large errors have to be expected if fatigue is estimated on an individual level and with high temporal resolution.

I. INTRODUCTION

Both distracted and fatigued drivers cause crashes. It is believed that both causations are underreported, since in many cases there is no evidence of driver distraction or fatigue at the scene of a crash. Moreover, it was observed that drivers are reluctant to admit such impairments and that their sense of responsibility decreases during fatigue. Therefore, the detection of driver fatigue and distraction by in-car FMT systems is urgently needed and will help to overcome these problems [1]. Hypovigilance is a deficit in vigilance and leads to a decreased ability to maintain long-term attention. Fatigued drivers sometimes fail to perceive and interpret unexpected, relevant changes in the environ-

ment and / or lack the ability to make effective decisions as well as perform precise motor movements.

The two major causes of hypovigilance are central fatigue and task monotony. But, it is well known that several other factors influence driver hypovigilance. It is a complex issue with several facets. It depends for example on time-of-day due to the circadian rhythm, on time-since-sleep (long duration of wakefulness), on time-on-task (prolonged work), inadequate sleep, and accumulated lack of sleep. The last two factors may be caused by pathological sleepiness due to diseases, like sleep apnea or narcolepsy, or may be caused by intentional sleep loss due to prolonged time awake. Moreover, there are also psychological factors influencing the actual level of vigilance, e.g. motivation, stress, and monotony. The latter is believed to play a major role during driving, because especially long-distance highway driving is in most situations a simple lane-tracking task with a low event rate and therefore deactivation may occur. Moreover, vigilance is considered as a psychophysiological variable not always decreasing monotonically during driving. It shows slow waxing and waning patterns, which can be observed in driving performance and repeatedly self-reported sleepiness.

Different FMT have been proposed to detect driver fatigue automatically, using a variety of fatigue related measures. In this paper, three video-based warning systems as well as electrophysiological signals were investigated.

To demonstrate this, two different procedures were applied. First, the outcome of the video-based FMT systems was compared to two fatigue measures which are independent of both approaches. The self-report of sleepiness on the Karolinska sleepiness scale (KSS) [2] acted as a subjective fatigue measure, and the driving performance in terms of standard deviations of lateral position in lane (SDL) [3] acted as an objective fatigue measure.

Secondly, video-based and electrophysiology-based approaches were compared directly using discriminant analysis. This was realized by recording both types of signals simultaneously during overnight driving simulations, and by performing the same kind of signal processing. The independent fatigue measures (KSS, SDL) were used to divide the data set in two classes: mild and strong fatigue. As a last step, discriminant analysis was performed, and the mean error rates for both the video-based and electrophysiology-based approaches, were compared.

Several authors who have investigated driving variables

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and / or driver behaviour variables, under different levels of fatigue, utilized descriptive statistics or linear discriminant analysis of time-domain features. Because drivers perceive a lot of heterogeneous information and experience changing road scenes, their driving behaviour is extremely situation specific. Moreover, the internal state of the driver is influenced by complex regulatory processes. This leads to non-unimodal distribution densities of the measured variables in the feature space. Modern computational intelligence methods like SVM are known to deal with this. They allow adapting nonlinear discriminant functions, and regularizing them over a wide range from strong locality to complete globality [4]. Another advantage is that they are non-parametric methods. Therefore, they overcome limitations of parametric statistics. This is important if effects are expected to vary from individual to individual.

The investigated FMT devices were all video-based. The data file stored on the host computer included results from the application of several variables such as time series of lid gap, blink duration, and pupil diameter. However, only the PERCLOS variable was utilized to measure driver fatigue. PERCLOS is defined as the percentage of time during which the pupils are covered by the eye lid by more than 80% of their area [5]. Thus it is an integral measure of prolonged eye closures. Typical integration intervals to calculate PERCLOS are between 1 and 5 minutes. Vendors of the three FMT agreed to our testing protocol, and provided integration assistance.

II. METHODS

A. Experiments

16 participants drove two nights (11:30 p.m. – 8:30 a.m.) in our real car driving simulation lab. One overnight experiment comprised of 8 x 40 min of driving. EEG (FP1, FP2, C3, Cz, C4, O1, O2, A1, A2) and EOG (vertical, horizontal) were recorded at a sampling rate of 256 Hz. PERCLOS as another oculomotoric measure was recorded independently utilizing the three FMT devices (P_A , P_B , P_C). As it turned out, there were no large differences in their technical principles. All were based on video analysis, utilized modulated infra-red light, and slightly different data rates (range 30 – 120 Hz). They all output several variables of eyelid and head movements, but utilized only PERCLOS as a measure of fatigue.

Also several variables of driving simulation, like steering angle and lane deviation were recorded at a sampling rate of 50 Hz. The driving performance in terms of standard deviations of lateral position in lane (SDL) [3] acted as an objective fatigue measure. The KSS, as mentioned above, is a standardized, subjective, and independent measure of hypovigilance on a numeric scale between 1 and 10. KSS was asked at the beginning and after finishing driving. During driving only relative changes in percent of the full

range were asked because subjects are more aware of relative than of absolute changes.

B. Data Analysis

For this study we analyzed the recorded data in two different ways. First, correlation analysis was performed between measured PERCLOS data of the three devices and the two independent fatigue measures (KSS, SDL). The PMCC was estimated.

Secondly, for the assessment of PERCLOS data as a fatigue measure in comparison with EEG/EOG, we performed nonlinear discriminant analysis. For this purpose, all signals under consideration (P_A , P_B , P_C , EEG/EOG) were transformed to the spectral domain, and the logarithmic power spectral densities (log PSD) were estimated by the modified periodogram method. Log PSD values of all signals were averaged in spectral bands. In case of EEG/EOG signals, 1.0 Hz wide bands and a range of 1 to 23 Hz turned out to be optimal, whereas in case of PERCLOS signals 0.2 Hz wide bands and a range of 0 to 4 Hz were optimal.

The independent fatigue measures (KSS, SDL) were both used to divide the whole data set into the two classes ‘mild fatigue’ (class 1) and ‘strong fatigue’ (class 2). This was necessary to get labels for discriminant analysis (classification). For this, histograms of all measures were computed and divided in two subsets of equal size. This was accomplished based on 3 minutes data segments.

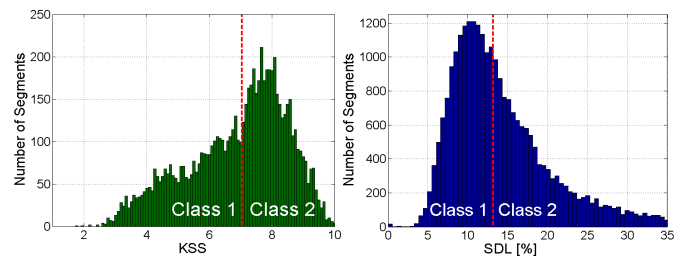


Fig. 1. Histogram of the subjective (KSS, left) and the objective (SDL, right) measure of driver fatigue. An arbitrary threshold (vertical line) was defined to have two classes (class 1: mild fatigue, class 2: strong fatigue).

The threshold parameter for the subjective measure was selected at KSS = 7.0 (Fig. 1a). The histogram indicates a slightly right-skewed distribution which confirms the success of the experimental design in inducing strong sleepiness. The same procedure was applied to the objective fatigue measure SDL (Fig. 1b). Subjects who performed lane tracking with standard deviations lower than 13.0% with respect to the lane width were defined to be in the state of mild fatigue. In our driving simulation system, the lane width was set to 5.0 m. This means that subjects in the mild fatigue state performed with standard deviations lower than 0.60 m. Lane tracking with standard deviations greater than 0.60 m was referred to the state of strong fatigue. Lateral positions in

lane greater than 100% with respect to the lane width were also exhibited, but were handled by the simulation system as an accident and led to a re-start procedure of the driving simulation.

III. RESULTS

First, the PERCLOS measures of the three FMT devices (P_A , P_B , P_C) were considered at relative high temporal resolution, all data were analyzed within each session (Fig. 2.). In non-overlapping 3 min intervals, mean and standard deviations of all 5 variables were calculated. Averaging across subjects was performed. This resulted in 13 mean values for each variable in each session. For comparison, all measures were scaled equally by setting the minimum to zero and the maximum to one for each measured time series (min-max scaling). The mean values of all measures, especially KSS, were influenced by time-on-task effects during all driving sessions. Calculations revealed fluctuations of the FMT variables which are not easily explainable by performance (SDL) or sleepiness (KSS) measures. In consequence, correlation coefficients broke down considerably. Even in sessions 1 to 4, where mild and mid-range sleepiness was rated, correlations diminished. Only sessions 5, 6, 7 displayed high correlations. Overall, the study showed large inter-individual difference in all measures (Fig. 2, standard errors).

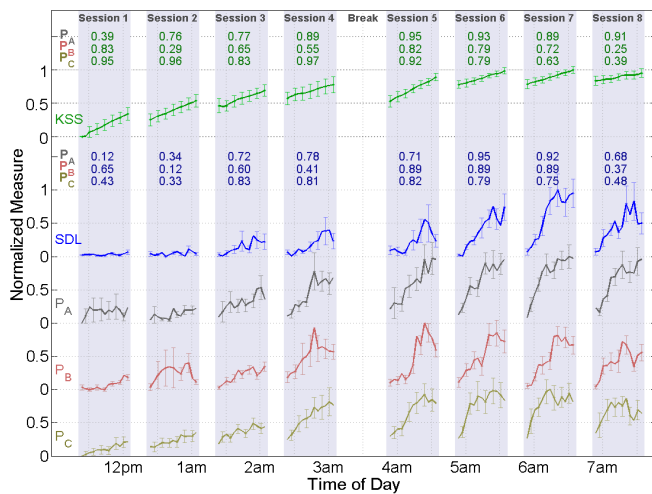


Fig. 2. Mean and standard deviations of the driver fatigue (KSS, green), driving performance measure (SDL, blue) and FMT output variables (P_A , P_B , P_C). All variables were averaged over 3 min intervals and across all subjects. PMCC between KSS/SDL and P_A , P_B , P_C for each session are listed.

To explain these large inter- and intra-individual differences, 3 minute segments of the PERCLOS time series $P_A(t)$, $P_B(t)$, and $P_C(t)$ were analyzed on a more general level using pattern recognition techniques combined with methods of computational intelligence. For simplicity, it was only

asked how well they may discriminate between mild and strong self-rated sleepiness, and alternately, how well they discriminate between high and low lane tracking performance. In addition, the same time series analysis was performed on EEG/EOG recorded simultaneously to PERCLOS. Under the assumption that sleepiness, as well as performance decrements, are originated in the central nervous system, the EEG is a more direct measure than EOG and PERCLOS. Despite this, all signals were covered by noise and by many other signal components not tightly related to sleepiness and performance decrements. Therefore, it is an open question as to which signal provides the best ability to detect driver fatigue.

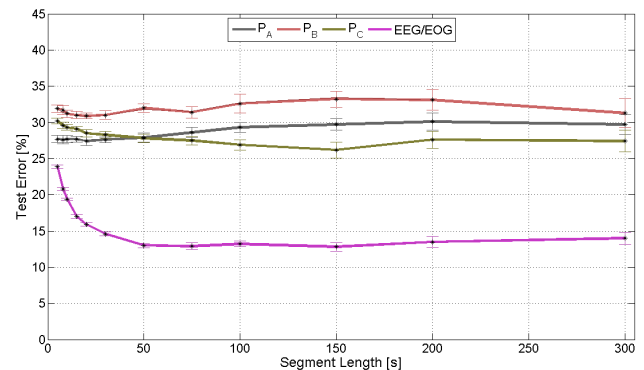


Fig. 3. Results of nonlinear discriminant analysis for the case of subjective ratings of sleepiness (KSS) as an external criterion of driver fatigue. Mean and standard deviations of the test error of classification are plotted versus the width of the signal analysis interval (length of data segments). EEG/EOG based data analysis performed much better than the PERCLOS based ones.

The discriminant analysis between mild and strong fatigue of the three PERCLOS signals yielded relatively high errors between 26% and 34% (Fig. 3). The errors remained in this range independent of the segment length. The same analysis of EEG/EOG resulted in substantially lower errors of about 13% when the segment length exceeds 50 seconds. Note that with increasing segment length, the number of segments and in consequence the number of feature vectors is decreasing. Therefore, with larger segment length the discriminant analysis becomes less complex and the statistical accuracy decreases. The first case should lead to decreasing error rate whereas the latter should lead to larger standard errors. Fig. 3 shows only the latter effect. Apparently, the reduced complexity due to data reduction is compensated by increasing complexity of the sampling distribution of the feature vectors, at least for segment length greater than 50 seconds.

If the same discriminant analysis is performed using driving performance (SDL) instead of sleepiness ratings (KSS) as class label, then higher errors (Fig. 4.) for all PERCLOS variables (range 27 - 41%) and lower errors for EEG/EOG (11% at 150 s) were found. For all signals, errors

decreased with increasing segment length. For both labels (KSS, SDL) the differences between the FMT devices are roughly the same. Device A performs slightly better for small segment length, whereas device C performs slightly better for large segment length. The results of EEG/EOG analysis confirm our previous findings [6].

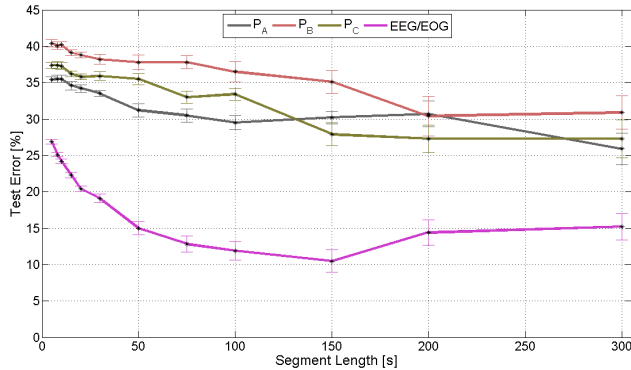


Fig. 4. Results of nonlinear discriminant analysis for the case of driving performance (SDL) as an external criterion of driver fatigue. Mean and standard deviations of the test error of classification are plotted versus the width of the signal analysis interval (length of data segments). EEG/EOG based data analysis performed much better than the PERCLOS based ones.

IV. CONCLUSIONS

The aim of the present study was to evaluate commercially available devices for driver fatigue monitoring. FMT devices were investigated in an overnight driving simulation study to evaluate their accuracy. One may make an objection to the lab study instead of a field study. However, in the field it is not possible to investigate such high levels of sleepiness and performance decrements. Furthermore, lab studies should be a first step for practical FMT evaluation. If devices fail under lab conditions then it is likely that they will fail under the much more complex influences of field conditions.

Subjectively rated sleepiness (KSS) and objectively assessed driving performance in terms of standard deviation of lateral position in lane (SDL) indicate increased deteriorations throughout the night. These two values were used as independent labels to evaluate the PERCLOS based fatigue measures of each of the three video-based FMT devices. Correlations between KSS, SDL and the PERCLOS measures P_A , P_B , P_C varied largely between different sessions (range 0.12 - 0.92) as well as between the three FMT devices. Clearly, this is not sufficient for FMT as a tool to prevent accidents in the field.

Therefore, the general ability of the PERCLOS measure of all three FMT devices to distinguish between mild and strong fatigue was of particular interest. As a benchmark a combination of EEG/EOG instead of PERCLOS as input signals were used in the same framework of data analysis. It has been shown recently that the combination of EEG and EOG is an excellent measure of strong fatigue when modern computational intelligence methods were utilized [6]. The

resulting mean error of discrimination between mild and strong fatigue was 13% and 10% for KSS and SDL labels, respectively, when EEG/EOG were analyzed. On the other hand, discriminant analysis of all three PERCLOS signals provided mean errors between 26% and 32% for KSS and between 26% and 42% for SDL labels, respectively. In contradiction to the results of Dinges [7], this shows that PERCLOS seems to carry less information on driver fatigue than EEG combined with EOG.

Similar conclusions were reached by Johns [8]. He pointed out that under demands of sustained attention, some sleep-deprived subjects fall asleep while their eyes remain open. Unfortunately, PERCLOS does not include any assessment of eye and eye lid movements. Important dynamic characteristics which are widely accepted, such as slow rolling eye movements, reductions in maximal saccadic speed, or in velocity of eye lid reopening, are ignored. Their spectral characteristics were picked up in our study through EOG and may account for the far better results of EEG/EOG data fusion presented here. Note, that highly dynamical alterations are better reflected by EOG and by eye-tracking signals than by integral measures like PERCLOS. Nevertheless, adaptive signal analysis of EEG/EOG in combination with computational intelligence methods resulted in highest detection performance.

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